

CSA1501 – Assessment Tool 2

Scenario-Based Questions – Detailed Answers

Question 1

A media company migrating its video streaming platform to the cloud requires a highly scalable architecture capable of supporting millions of concurrent users while maintaining minimal buffering. A suitable architecture begins with a global Content Delivery Network (CDN) that caches videos on edge servers located close to users. Requests first reach the CDN, reducing latency and minimizing traffic to the origin servers. A global load balancer distributes user requests across multiple cloud regions based on proximity, health, and server availability.

The application servers run inside virtual machines or Kubernetes containers behind regional load balancers. Auto-scaling policies automatically increase or decrease the number of servers based on CPU utilization, bandwidth consumption, and request rate. During major events or new content releases, additional instances are launched automatically. When traffic decreases, unnecessary instances are terminated, reducing operational costs.

Object storage services store video files because they provide high durability, scalability, and low maintenance. Frequently accessed content is cached at edge locations to improve playback performance. Databases should use managed cloud database services with read replicas for handling large numbers of user requests.

Cost optimization can be achieved by using reserved instances for predictable workloads, spot instances for background processing, lifecycle policies for rarely accessed content, and intelligent storage tiers. Compression and adaptive bitrate streaming also reduce bandwidth costs while improving user experience.

Monitoring tools continuously track CPU usage, response time, cache hit ratio, streaming quality, and network latency. Automated alerts enable administrators to resolve issues quickly. A practical example is Netflix, where cloud scalability, CDN caching, load balancing, and monitoring ensure uninterrupted streaming for millions of viewers worldwide.

Question 2

A fintech startup deploying containerized applications across multiple cloud providers requires consistency, portability, and simplified management. The best solution is to package every application as Docker containers and manage them

using Kubernetes. Kubernetes provides a common orchestration platform regardless of whether the application runs on AWS, Azure, Google Cloud, or a private cloud.

Infrastructure should be defined using Infrastructure as Code (IaC) tools such as Terraform so identical environments can be deployed across providers. Continuous Integration and Continuous Deployment (CI/CD) pipelines automate application building, testing, and deployment.

Service discovery is handled by Kubernetes Services and internal DNS. Applications communicate through service names rather than fixed IP addresses, allowing containers to move freely without affecting connectivity. Kubernetes also provides automatic load balancing, rolling updates, self-healing, and horizontal pod auto-scaling based on CPU or custom metrics.

A multi-cloud strategy improves business continuity because workloads can continue running even if one cloud provider experiences an outage. It also reduces vendor lock-in and allows organizations to select the most cost-effective or highest-performing services from different providers.

However, multi-cloud deployments introduce challenges such as increased operational complexity, higher networking costs, inconsistent security policies, and more difficult monitoring. Organizations should implement centralized monitoring, identity management, and security controls across all cloud platforms.

Overall, Kubernetes combined with Docker, Terraform, CI/CD pipelines, and centralized monitoring provides a reliable, scalable, and portable multi-cloud solution suitable for financial applications.

Question 3

An e-learning platform experiencing accidental data deletion requires a strong backup and disaster recovery strategy. Cloud-native backup solutions should automatically create scheduled backups of databases, storage buckets, virtual machines, and application configurations. Backup copies should be stored in multiple regions to protect against regional failures.

Object storage versioning should be enabled so deleted or modified files can be restored to previous versions. Cross-region replication continuously copies important data to another cloud region, ensuring availability during disasters.

Recovery Time Objective (RTO) is minimized by maintaining standby infrastructure

using Infrastructure as Code and automated recovery scripts. Recovery Point Objective (RPO) is reduced through frequent snapshots, continuous replication, and scheduled database backups, ensuring minimal data loss.

Business continuity requires regular backup testing, disaster recovery drills, documented recovery procedures, automated failover mechanisms, and continuous monitoring. Identity and access controls must protect backup repositories from unauthorized modification or ransomware attacks.

For example, if an instructor accidentally deletes important course content, versioning allows immediate restoration. If an entire cloud region fails, replicated backups stored in another region enable rapid recovery, ensuring students continue accessing learning materials with minimal interruption.

Question 4

A government agency handling sensitive and non-sensitive information should adopt a hybrid cloud architecture. Sensitive citizen records remain within the on-premises data center to satisfy compliance requirements, while public websites, email services, analytics, and collaboration platforms operate in the public cloud.

Secure connectivity between environments should be established using VPN tunnels or dedicated private connections such as AWS Direct Connect or Azure ExpressRoute. Firewalls, network segmentation, and intrusion detection systems protect communication between environments.

Identity management should be centralized using directory services integrated with cloud Identity and Access Management (IAM). Single Sign-On (SSO), Multi-Factor Authentication (MFA), and role-based access control ensure that users receive only the permissions necessary for their responsibilities.

Encryption must protect data both during transmission using TLS and while stored using AES-256 encryption. Encryption keys should be managed securely through dedicated key management services.

Managing hybrid cloud environments introduces challenges such as increased operational complexity, maintaining consistent security policies, synchronizing data, monitoring multiple platforms, and meeting compliance requirements. Centralized monitoring and automation help simplify administration while maintaining security and regulatory compliance.

Question 5

A SaaS provider requiring continuous availability should deploy its application across multiple cloud regions. Each region contains identical application servers, databases, storage, and networking components. A global load balancer directs users to the nearest healthy region and automatically reroutes traffic if one region becomes unavailable.

Databases should use cross-region replication with synchronous replication where low latency permits and asynchronous replication for distant regions. Object storage should also replicate data automatically across multiple regions. Automated failover mechanisms continuously monitor service health and switch traffic to healthy regions without manual intervention.

Application servers should run behind regional load balancers with auto-scaling enabled to handle changing workloads. Stateless application design allows new instances to start quickly during demand spikes.

Comprehensive monitoring is essential. Cloud monitoring tools track CPU utilization, memory usage, application latency, database performance, network health, and regional availability. Alerts are generated immediately when abnormal behavior occurs. Dashboards provide real-time visibility, while log analysis supports rapid troubleshooting.

A practical example is Microsoft 365 or Salesforce, where users continue accessing services even if one data center experiences an outage. Multi-region deployment, replication, failover, and continuous monitoring together ensure high availability, fault tolerance, business continuity, and excellent customer experience.